

MICHAEL LEI

michael.lei@duke.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Duke University

Durham, NC

B.S. Electrical and Computer Engineering, B.S. Computer Science, GPA: 4.0/4.0

May 2028

- Relevant Coursework: Software Design, Algorithms, Databases, OS, Networking, Machine Learning, Computer Architecture

EXPERIENCE

Forewarned

May 2026 – Present

Software Engineer

Durham, NC

- Engineered distributed microservices that normalize thousands of daily block events from Palo Alto firewalls, AWS security groups, Azure network policies, and edge routers into a unified event schema.
- Shipped a retrieval service that embeds newly published CVEs on ingest and reranks blocked source IPs by matched vulnerability recency, prioritizing operator response on critical threats.
- Built an operator dashboard surfacing priority-ranked attacker queues and cross-device campaign views over previously siloed firewall, cloud, and router event streams.

VSee

May 2025 – Present

Software Engineering Intern

Sunnyvale, CA

- Engineered dockerized backend microservices for clinical de-identification and ICD / CPT medical coding with Kubernetes-style health probes and Prometheus metrics, reducing physician charting time by **93%** in production.
- Extended VSee's on-prem LLM platform with 8 MCP server integrations, authoring a custom Elasticsearch analytics MCP server, exposing internal docs, repos, and product usage data, increasing platform usage by **17%**.
- Shipped a PR review service triggered by Bitbucket webhooks that posts line-cited issues via Code Insights reports, inline annotations, and PR comments, running on **100+ production PRs across 3 repos**.

Chilkoti Group

Summer 2025

Software Engineer

Remote

- Automated per-slide OpenCV parameter tuning with a coarse-to-fine grid search, cutting manual tuning from **20 minutes to 2 minutes per slide**.
- Designed a brightness-contrast scoring metric for the search, enabling parameter selection without labeled ground truth.
- Productionized the lab's Jupyter-based CV workflow into a React UI for interactive slide annotation.

Duke University

Fall 2024

Software Engineer

Durham, NC

- Engineered a flagpole control system pairing a browser UI with C++ firmware on ESP32 devices, targeting a manual process that consumed hundreds of staff-hours per year.
- Architected a distributed MQTT control plane where new flagpoles auto-onboard via retained topics and a single broadcast topic fans commands across the full device fleet.

PROJECTS

DukeDine | *Python, React, PostgreSQL, AWS ECS, OpenTofu* | [Demo](#)

- Designed and shipped a full-stack nutrition and strength-training platform serving **100+ active Duke users**, deployed on AWS ECS Fargate with OpenTofu-managed infrastructure and CI/CD.
- Built the personalization backend on AWS Bedrock with a retrieval layer over embedded user training logs, validating cited dates and quotes against source records before responses are returned.

Bet308 | *Java, JavaFX* | [GitHub](#)

- Architected a data-driven card-game engine with an LLM-assisted authoring environment that compiles natural-language prompts into playable game JSON, shipping **7 games on a single interpreter**.
- Led architecture on an **8-person team** through 6 weeks of agile development, running one-week shippable sprints.

FireWatch | *Swift, CoreML, DJI Mobile SDK*

- Built a wildfire detection system, developing a Swift iOS app that streams a drone's live video feed through an on-device CoreML model at **~95% test accuracy**, enabling early intervention.
- Won **U.S. Division Semi-Finalist** honors at the China-US Young Maker Competition.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript / JavaScript, Java, Swift, C++, SQL

Web & Backend: Microservices, Distributed Systems, System Design, REST APIs, WebSockets, Webhooks, Event-Driven Pipelines, Caching, FastAPI, React, Node.js, PostgreSQL, Redis, JWT, OAuth

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, GitHub Actions, Bitbucket Pipelines, OpenTofu, Observability, Linux, Bash, Git

AI / LLM Engineering: RAG, MCP, AI Agents, prompt engineering, fine-tuning, embeddings